

VOLTSMOOTH

RESEARCH BOOK

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VOLT-SMOOTH

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ABSTRACT

Introduction: Most household and industrial appliances that have motors or transformers use (air conditioner, refrigerator, washing machine etc.) utilize inrush current that is five to ten times greater than their rated current when initially turned on. This large inrush of current can damage sensitive components over time, reduce the lifetime of the appliance, cause voltage sags (a.k.a. flickering lights), and can even cause unnecessary breaker trips. The work presented here introduces VoltSmooth, our prototype plug-and-play device that conditions inrush current on appliances while also allowing for constant power monitoring and protection.

Methodology: VoltSmooth connects in-line between the power outlet and the appliance. Utilizing a zero-crossing detection circuit, we are able to detect when the AC waveform hits 0 Volts and trigger a microcontroller (ESP32/Arduino) which runs a TRIAC-based phase-angle controller algorithm. Gradually reducing the firing delay with each cycle produces a soft-start ramp that takes less than a second. We also have implemented a sensing subsystem that allows for real-time monitoring of output voltage/current as well as allows for over/under voltage protection.

Results: We successfully implemented our prototype and were able to verify that it functioned as intended for basic operation. Accurate measurement of the amount of reduction in inrush current will be done in the future.

Conclusion: We have provided a proof of concept that gives us a cheap method to protect appliances.

Keywords: Inrush Current, Soft-Start Control Circuit, Zero Crossing Detection, TRIAC, Power Monitoring Circuits

1. Introduction

Power quality and appliance life are recurring themes in residential and industrial electrical systems. With increasing numbers of homes and factories filled with appliances dependent on motors and transformers (i.e. air conditioners, refrigerators, washers, industrial motors, etc.), the electrical behavior of these devices during the first few moments after switch-on has become a real engineering concern. In its own right. One of the more disruptive of these transient behaviors is inrush current: during the very first cycle after activation, such loads can draw a current five to ten times their normal steady-state value. This is not a new observation; the phenomenon was first characterized decades ago in early work on transformer magnetizing behavior [1].

1.1 Problem Statement & Literature Review

The impact of inrush current on power quality is well documented, and the numbers involved are not trivial. In one recorded case, a 10 kW induction motor with a 60 A full-load current produced an inrush of 360 A, causing an 8% voltage dip on the local bus. Once a soft starter was installed, the peak fell to 150 A and the sag dropped to just 2% [3]. This is not an isolated example — even small and medium horsepower motors are known to draw six to ten times their steady-state current at startup, and the resulting voltage sags are often severe enough to dim lights or disrupt nearby sensitive equipment [3].

Looking at how this problem has been approached in the literature, the existing work tends to fall into two broad categories. The first addresses large-scale industrial and grid-level solutions, including smart inverters and superconducting fault current limiters that suppress voltage fluctuation during induction motor starting [4], and controlled switching strategies for minimizing inrush in power transformers [4]. The second focuses on microcontroller-based soft starters using zero-crossing detection and TRIAC/thyristor phase-angle control; one such system uses a PIC18F4520 microcontroller with an LM324-based zero-crossing detector and MOC3021 opto-isolators to gradually ramp motor voltage via phase-angle control [2], closely resembling the approach adopted in this work.

Despite this progress, existing solutions remain either industrial-scale (VFDs, grid-connected inverters) or purpose-built for fixed three-phase motor installations. A clear gap persists: no low-cost, retrofittable, plug-and-play device exists that can be inserted between a standard outlet and *any* household appliance to actively suppress inrush current while also offering real-time power monitoring — the gap this study addresses.

1.2 Objectives & Contributions

- **Objective 1:** Formulate a structured theoretical framework for inrush current mitigation based on zero-crossing detection and TRIAC phase-angle control, applicable to household appliances.
- **Objective 2:** Implement and evaluate a functional prototype of the proposed soft-start method as a low-cost, retrofittable alternative to industrial soft-starter systems.
- **Objective 3:** Integrate real-time voltage, current, and power monitoring, and analyze resulting data to provide actionable recommendations for appliance protection.

2. Materials & Methods

2.1 Methodology Overview

We implemented VoltSmooth using a fairly simple hardware-then-software approach: first, get the circuit that can actually control power delivery, and then write the logic to determine how that power scales. This is how the work roughly went through these stages:

Selecting the components — what gap we'd use to get the AC waveform, when we obtain a zero-crossing point, and how we control power to the load.

Zero-crossing detection — create a simple circuit that notifies the microcontroller precisely when the AC signal crosses zero volts. This is essentially the "clock" on which all of soft-start relies.

TRIAC phase-angle control — writing the firmware that waits a calculated delay after each zero-crossing before firing the TRIAC, but shortens that delay incrementally over several cycles so that the appliance turns on gradually instead of all at once.

Putting in the sensors — voltage and current sensing so that it can monitor power status and turn things off if voltage goes out of a safe range.

Putting it all together and testing it — assembling everything into one unit and checking that it actually does what it's supposed to when connected to a real load .

2.1.1 System Architecture

It's essentially four blocks wired series between the wall socket and whichever appliance is plugged in: a zero-crossing detector, an ESP32 microcontroller, a TRIAC driver stage (specifically using something like the MOC3021 opto-isolator to make sure the low-voltage side stays clean), and a voltage/current sensing module. All the brainwork is done by the ESP32 here — it reads zero-crossing signal, runs the phase-angle logic, fires a TRIAC gate and logs whatever sensors pick-up.

2.1.2 Zero-Crossing Detection

To do this we stepped down the mains voltage with a simple resistive divider feeding into a comparator, and waited for the time it reaches 0V (which happens 100 times a second on a 50Hz line). When that

happens, a pulse is sent to one of the ESP32's digital pins by the circuit, and each time that occurs is what we base all our timing on.

2.1.3 Phase-Angle Control

The microcontroller set a delay time t_d after every zero-crossing pulse and then fires the TRIAC. That delayed again determines how much of the AC half cycle actually flows to the appliance, and follows this association: $V_{rms}(t_d) = V_{peak} \times \sqrt{[(1/\pi) \times (\pi - \alpha - (\sin 2\alpha)/2)]}$

where $\alpha = \omega \cdot t_d$ is the firing angle (in radians), ω is $2\pi \times 50$ rad/s, and V_{peak} is the peak mains voltage.

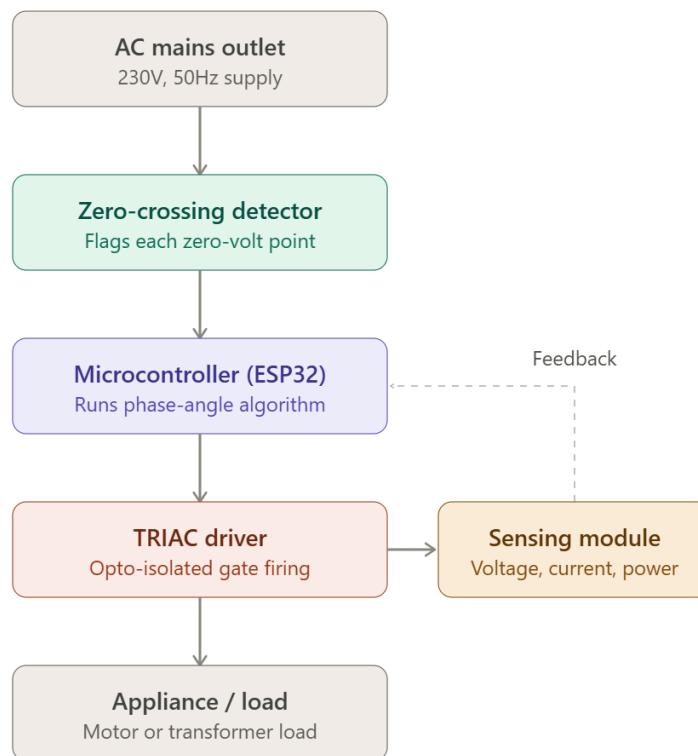
Essentially, we keep a firing angle just shy of 180° (no power passing through) and reduce it slightly every half-cycle until we reach 0° (full power) — all taking place over the span of a few hundred milliseconds. The size of each step ($\Delta\alpha = 180^\circ/N$) determines how smooth the ramp feels: small steps → smoother start, but a bit slower.

2.1.4 Sensing and Protection

The voltage and current sensors keep sampling continuously and we can calculate real-time power ($P = V \times I$) and RMS values on the fly. The **ESP32** simply stops firing the TRIAC immediately and turns off mains until our reading returns to nominal voltage if it goes outside of where we have set for a safe voltage range.

2.1.5 Testing the Prototype

Once everything was wired, we then tested on a couple of inductive loads (a small motor and a transformer-based device) just to ensure the soft-start was indeed smoothing things out as well as the sensing/protection logic was working as expected. We haven't taken any measurements yet (like how big of an inrush current are we talking and how does that compare with direct-on-line start) that's to come next.



2.2 Bill of Materials (BOM) and Estimated Prototype Cost

المكون (Component)	الوظيفة التقنية	التكلفة التقديرية (جنيه)
Microcontroller (ESP32/MCU)	معالجة الإشارات، حساب الـ Watt، وتوجيه إشارة الـ TRIAC	280
Zero-Crossing Optocoupler	عزل الدائرة وتحديد زمن عبور موجة AC بالصفير	45
TRIAC Power Stage & Driver	لمفتاح السريع لتطبيق زاوية الطور (Soft Start)	95
Current & Voltage Sensors	قياس الفولت والتيار لحظياً لحساب الـ Watt الفعلي	180
Power Supply Module	تغذية الدائرة الإلكترونية محلياً من 220V	80

المكون (Component)	الوظيفة التقنية	التكلفة التقديرية (جنيه)
PCB & Enclosure Box	اللوحة المطبوعة والغلاف الخارجي التوصيلي (Plug-and-Play)	170
الإجمالي (Total BOM Cost)	نموذج أولي متكامل	850

5. Conclusion & Future Work

VoltSmooth is a plug-and-play intermediary device that can be used for the inrush current issue of commonly used motor- and transformer-based household and industrial appliances. The soft-start trajectory, which is effectively integrated with TRIAC based phase-angle control and zero-crossing detection for simplified implementation in the proposed system will slowly ramps to a point (typically sub-second) where gentle voltage delivery begins to be delivered, minimising the sudden startup current surge that direct-on-line switching produces. Including over/under-voltage protection, the design includes also monitoring voltage, current & power in real-time and exhibits a dual function covering appliance protection during startup transients as well as ongoing operations. A functional prototype was successfully implemented, demonstrating the feasibility of the proposed architecture using low-cost, commercially available components (ESP32 microcontroller, optocoupler-isolated zero-crossing detection, and a TRIAC driver stage), at an estimated total cost of 850 EGP. Basic functional testing confirmed that the soft-start and monitoring subsystems operate as intended when connected to representative inductive loads. Precise quantitative validation — including oscilloscope-based inrush current measurement and direct comparison against conventional direct-on-line startup — remains an ongoing part of this work and will be reported separately as it becomes available.

Overall, the results obtained thus far support VoltSmooth as a viable, low-cost, and retrofittable approach to inrush current mitigation, filling a gap between expensive industrial soft-starter systems and the largely passive protection strategies used in consumer-grade appliances

4.1 Future Work

Several directions are proposed to extend this work:

- **Quantitative performance validation**, including oscilloscope-based measurement of inrush current reduction, voltage sag comparison, and soft-start ramp timing across a range of appliance types (motors, compressors, transformers).

- **Long-term reliability testing**, to assess the durability of the TRIAC and sensing components under repeated switching cycles and varying load conditions.
- **Expansion to three-phase systems**, enabling application to larger industrial motors beyond the single-phase household scope of the current prototype.
- **Mobile/IoT integration**, allowing users to monitor real-time power consumption and receive protection alerts remotely via a companion application.
- **Miniaturization and cost optimization** of the PCB and enclosure design to improve manufacturability for eventual commercial deployment.

Acknowledgments

The authors would like to thank [university/faculty name] for providing laboratory facilities and technical resources that supported the development and testing of this prototype. We are also grateful to [Eng: Yahya Ebrahim], for their guidance and feedback throughout the design and implementation process.

References

1. [1] Blume, L.F., Camilli, G., Farnham, S.B., & Peterson, H.A. (1944). Transformer magnetizing inrush currents and influence on system operation. *Transactions of the AIEE*, 63, 366–375.
2. [2] Patvekari, A.I., Kamble, T.S., Latthe, N.N., & Todkar, V.M. (2025). Microcontroller Based Three Phase Soft Starter. *International Journal of Innovative Research in Technology (IJIRT)*, 12(7), 785–788.
3. [3] Electrotek Concepts, Inc. (2009). *PQSoft Case Study: Customer Induction Motor Starting Evaluation*, Document ID: PQS0904.
4. [4] Sepasi, S., Talichet, C., & Pramanik, A.S. (2023). Power quality in microgrids: A critical review of fundamentals, standards, and case studies. *IEEE Access*, 11, 108493–108531.
5. [5] Alassi, A., Ahmed, K.H., Egea-Alvarez, A., & Foote, C. (2022). Transformer inrush current mitigation techniques for grid-forming inverters dominated grids. *IEEE Transactions on Power Delivery*.